

Standard Operating Procedure

SOP-114 — Mechanical Seal Replacement, AF-4500 Series

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Purpose

This procedure defines the steps for safely replacing the mechanical seal on AF-4500 series pumps, whether during scheduled annual maintenance or in response to seal leakage.

Scope

Applies to all AF-4510, AF-4520, and AF-4530 units in field and shop service.

Required PPE

- Safety glasses
- Nitrile gloves
- Steel-toe boots
- Hearing protection if in a pump room

Required Tools & Parts

- Replacement seal kit RSK-4500 (model-specific)
- Atlas HydraSeal AH-220 lubricant
- Torque wrench, 10–150 Nm range
- Mechanical seal puller
- Dial indicator for coupling alignment check
- Lockout-tagout kit

Pre-Job Safety (Lockout-Tagout)

- 1 Notify operations that the pump will be taken out of service.
- 2 Isolate and lock out the motor electrical supply; verify zero energy state.
- 3 Close and lock out suction and discharge isolation valves.
- 4 Depressurize and drain the pump casing before proceeding.

Procedure

- 1 Confirm LOTO is in place and casing is fully drained.
- 2 Disconnect the coupling and remove the coupling guard.
- 3 Remove the bearing housing cover and slide back the rotating assembly per the IOM manual.
- 4 Extract the worn mechanical seal using the seal puller; avoid scoring the shaft sleeve.
- 5 Inspect the shaft sleeve for wear or scoring; replace if damage exceeds 0.1 mm.
- 6 Clean the seal chamber thoroughly, removing all old gasket material and debris.
- 7 Lightly lubricate the new seal's elastomer components per the seal kit instructions.
- 8 Install the new mechanical seal to the working length specified on the RSK-4500 data card.
- 9 Reassemble the rotating assembly and bearing housing cover.

- 10 Re-align the coupling using the dial indicator; confirm alignment is within 0.05 mm.
- 11 Refill the bearing housing with Atlas HydraSeal AH-220 to the sight-glass level.
- 12 Reinstall the coupling guard.
- 13 Remove lockout-tagout devices in reverse order of application.
- 14 Slowly open suction and discharge valves, prime the pump, and perform a function test per Section 5.2 of the IOM manual, checking for leaks at the new seal.

Post-Job Checklist

- No visible seal leakage after 15 minutes of running.
- Bearing housing temperature and vibration within normal ranges (Section 5.3, IOM manual).
- Coupling guard secured; work area cleaned; used seal disposed of per site waste procedure.
- Service record logged in Atlas FieldOps with old and new seal serial numbers.

Revision History

Rev	Date	Change
A	June 2024	Initial release
B	February 2026	Added shaft sleeve wear check; updated seal kit part number to RSK-4500

SYNTHETIC DOCUMENT — created for a portfolio project simulation.